Solution Architecture

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Team ID	G05
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The System is designed to automate the extraction and validation of information contained in uploaded documents.

Organizations often rely on documents such as identity cards, certificates, and regulatory records to verify claims made by individuals or organizations. Current workflows rely heavily on manual verification, which is slow, error-prone, and difficult to audit.

The proposed system introduces an automated verification pipeline that:

1. Accepts uploaded PDF documents
2. Extracts text and layout information
3. Detects verifiable fields using AI-based field detection
4. Maps fields to a canonical schema
5. Validates extracted fields using trusted external APIs
6. Displays verification results directly in the document viewer

The system provides **field-level verification with visual indicators**, allowing users to quickly determine whether document information is valid or requires manual review.

System Architecture Overview

The system follows a **modular service-oriented architecture** consisting of:

- Frontend user interface
- Backend API services
- Document processing services
- AI field detection services
- Validation connector layer
- Storage and audit systems
- External verification APIs

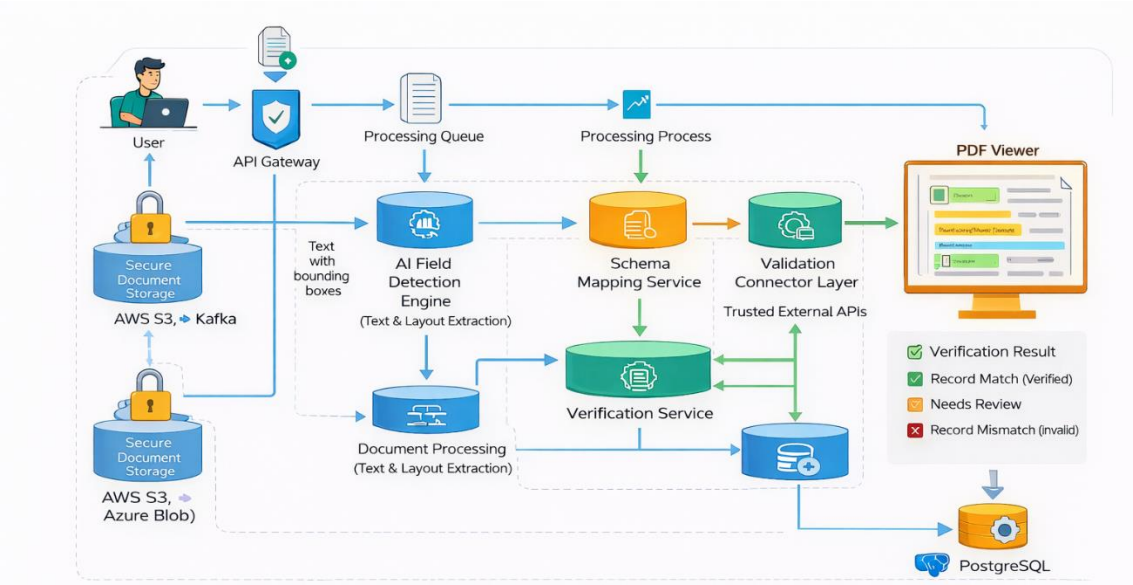
The architecture separates document ingestion, processing, and validation into independent services to ensure scalability and maintainability.

Component Architecture

Core System Components

1. PDF Upload Interface: Allows users to upload documents.
2. API Gateway / Backend Service: Manages requests and triggers processing pipelines.
3. Secure Document Storage: Stores uploaded documents with encryption and retention policies.
4. Processing Queue: Handles asynchronous document processing tasks.
5. Document Processing Engine: Extracts text and layout data.
6. AI Field Detection Engine: Detects key fields using AI-based analysis.
7. Schema Mapping Service: Maps extracted fields to document specific schema elements.
8. Validation Connector Layer: Sends extracted data to external validation APIs.
9. Verification Service: Determines final verification status for each field.
10. Audit Logging Service: Stores verification actions and manual overrides.
11. Verification Viewer: Displays documents with color-coded verification overlays.

Component Diagram



Data Flow Overview

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User
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PDF Upload Interface
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API Gateway
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Secure Document Storage
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 ▼
Processing Queue
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 ▼
Document Processing Engine
(Text + Layout Extraction)
 |
 ▼
AI Field Detection Engine
 |
 ▼
Schema Mapping Service
 |
 ▼
Validation Connector Layer
 |
 ├──────────► Trusted External APIs
 |
 ▼
Verification Service
 |
 ├──────────► Audit Logging Service
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PDF Viewer
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External Integrations

The system integrates with several external services for verification purposes.

Trusted External APIs

Examples include:

- Identity verification APIs
- Verified ID services
- Institutional verification systems

These services provide authoritative validation of extracted document fields.

AI Model Providers
AI models may be hosted internally or accessed via external services.
Examples:

- Google Gemini
- Azure AI services
- custom ML models (least priority)

These models assist in field detection and document understanding.

Cloud Storage Services
Cloud storage solutions may be used to store uploaded documents and logs.
Examples:

- AWS S3
- Azure Blob Storage